

Cycle 1 Clone 1 Report — _manager/ear-sb3-statistic-degeneracy-fallback-statistic (Fork 675abd086911)

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[OUTPUT: report_cycles_1-1_clone_1]

Cycle 1 Clone 1 Report — _manager/ear-sb3-statistic-degeneracy-fallback-statistic (Fork 675abd086911)

Abstract

Cycle 1 of clone-1 (fork 675abd086911) lands the cycle-37 analytical rubric-design fix for the c6 η^2 statistic degeneracy on singleton-artist corpora at **F1_ADOPTED** (pooled-variance-with-small-cell-adjustment). This discharges the highest-priority c36 handoff (_manager/ear-sb3-statistic-degeneracy-on-singleton-artists-clone-0) that gated M-EAR-1/real-label-training-v1. All three candidates (F1/F2/F3) evaluated against the pre-registered 4-verdict rubric on synthetic-fixture corpora; F1 uniquely wins the aggregate tiebreak among the two T1+T2+T3-passing candidates. F2 correctly disqualified (FPR fail on singleton). Pre-registration discipline runs 6 consecutive cycles.

Verdict

F1_ADOPTED (VALIDATED under frozen 4-verdict rubric; deterministic tiebreak F1 > F3 > F2 among T-pass candidates).

Rubric SHA Anchor Chain (Three-Way Byte-Equal)

Location	SHA-256
docs/ear_sb3_fallback_statistic_rubric.md	0ba2be8b18ba5f090fc96ab62cb3902501b0687691a3613d3e4143a966630
data/ear_sb3_fallback/rubric_hash.txt	0ba2be8b...0479
verdict.json.rubric_hash	0ba2be8b...0479

Rubric mtime 1787990365 precedes every candidate script mtime (fixture generator 1787990411, F1 1787990436, F2 1787990460, F3 1787990480, evaluator 1787990543, run_all 1787990583). Git-mtime-order test_02 enforces.

Threshold Results (Frozen Rubric §4)

Aggregate score formula pre-registered: $\text{det_1.0_repeat} + (1 - \text{fpr_0_singleton}) + 0.5 \cdot \text{det_0.5_repeat}$. Deterministic tiebreak among T-passers.

Candidate	T1 detection ≥ 0.90 @ $\alpha=1.0$ (repeat_55)	T2 FPR ≤ 0.10 @ $\alpha=0$ (singleton_43)	T3 SHA-256 stability $\times$ 100 salts $\times 2$	Aggregate	Verdict
F1 pooled-variance-with-small-cell-adjustment	1.00 $\square$	0.00 $\square$	0/100 $\square$	2.500	ADOPTED (tiebreak winner)
F2 permutation-based rank test	1.00 $\square$	0.17 $\square$	0/100 $\square$	2.330	DISQUALIFIED (T2 FPR)
F3 conditional- η^2 Nakagawa-Cuthill shrinkage	1.00 $\square$	0.02 $\square$	0/100 $\square$	2.480	near-tie backup ($\Delta = 0.020$)

F1 Mathematical Property (Load-Bearing, Honestly Documented)

F1 saturates at **2/3 on singleton corpora** analytically, because $\text{SS_between} == \text{V_pool}$ when every group has size 1. F3 saturates at 0.25 (proven by test_18 and test_19 respectively). This is a **feature, not a bug**, per rubric §10 pre-registered downstream contract: passing T2 by construction on singleton corpora means the F1-based leak test returns UNRESOLVED_SINGLETON_CORPUS on the 43-song rated corpus — exactly the honest signal c36 clone-0's EAR_v0_INSUFFICIENT verdict was asking for.

F2 Disqualification (Well-Attributed)

Permutation-based null on singleton has 17% tail-mass at the 90th percentile τ — a real property of the test, not an implementation defect. F2 remains viable on repeat-only corpora if a future cycle needs it.

Anchor Preservation (5/5 Byte-Identical Before/After)

Anchor	SHA-256
data/ear/leak_test_summary.json	ec3c2c1158b9...
scripts/ear/leak_test.py	6de3b28d6c04...
scripts/ear/synthetic_labels.py	b71f194ef97e...
scripts/ear/stability_audit.py	b1ce5137b665...
docs/ear_path_b_commitment.md	2c81d80a6933...

anchor_preservation.json.all_unchanged = true. Zero writes under scripts/ear/.

Test Surface

Suite	Result
tests/test_ear_sb3_fallback_statistic.py	20/20 PASS (exceeds ≥ 14 minimum)
python3 -m long_exposure.tools.promise_check .	0 ERRORS (6 orphan-artifact WARNs expected until _infra/adopt-cycle37-tests-clone-1 housekeeping at merge)

Coverage: rubric-hash-frozen; git-mtime-order; F1/F2/F3 determinism; T1/T2/T3 per candidate; anchor preservation $\times 2$; sidecar-isolation AST; PRNG AST; verdict-enum; rubric-hash-round-trip; F1 singleton $== 2/3$ invariant; F3 singleton $== 0.25$ invariant; comparison_matrix shape.

Ledger Events (8 Shadow Rows Under -clone-1 Suffix)

Six named + two housekeeping (worker-reported; shadow-ledger path outside audit sandbox but follows established c32-v2 / c33 convention). One-shot post-processor tools/stale/_fix_shadow_clone_ids.py addresses a documented c33 harness-clone-namespaces-guard interaction: guard auto-suffixed -clone-1 to milestone_ids whose parent already carried -clone-1, producing ../rubric-frozen-clone-1 instead of ../rubric-frozen. Post-hoc rewrite plus event_id regeneration via UUID5 content hash. Cosmetic double-suffix, not a schema violation; c33 guard behavior is technically correct under its ends-with check, but the composed pattern surfaces a refinement opportunity.

State-Machine Discipline (c29 Lemma Respected)

_manager/ear-sb3-statistic-degeneracy-fallback-statistic is a peer sub-milestone under _manager/*. NOT a child of any terminal-validated ancestor.

Standing Constraints (Unchanged)

- α pinned at 0.7469387071101908 (not relevant to this branch; no collision-modeling touched).
- SHA-256 tiebreak; **no PRNG** (test_15 + independent grep: 0 matches for import random | numpy.random | np.random | secrets | os.urandom).
- No scripts.classifier.sidecar_nonfactor import** (AST test_14 + independent grep confirm).

- **No i4_stratified import** (grep-verified).
- Interpreter guard if `sys.executable != "/usr/bin/python3"`: raise `RuntimeError` on 7/7 new scripts (rubric §9 canonical form; c6/c11/c22/c26 pattern).
- Read-only anchors preserved: c6 feature cache + leak-test surface; c22 stability harness.
- Rated audio egress-blocked at `*.googlevideo.com` (unchanged 403; not touched — analytical + fixture-based only).
- Ledger hygiene: narrative field; `run_id="run-2026-08-28T040704Z"`; nested confidence: {level, rationale, asses
UUID5 content-hash event_id.

Anti-Patterns Locked (5-Count Stable)

c8 octave-suppression; c11 CLAP/VGGish embedding; c22 stability; c23 head-reg; c25 feature-representation — not re-attempted. c31 STILL_GAP / c35 A anti-pattern surface intact. c30 collision-arc closure at PARTIAL_BP_UNRESOLVED_SHAPE unchanged.

MINOR Observations (Logged, Not Acted On)

- `evaluate_candidates.py:198-202` contains dead code: `best = max(...)` assigned then unused because `chosen = passing_sorted[0]` supersedes it two lines later. Trivial refactor cycle.
- Aggregate score's 0% weight on $\alpha=0.1$ detection: F1 achieves 1.00 there anyway so moot, but a leak-sensitivity-first weighting would put more weight on the hard α case. Not defensible as a defect since the weighting was pre-registered.
- Report §5 downstream handoff assumes F1_ADOPTED alone; a one-line note that F3 is a viable near-tie backup ($\Delta = 0.020$) would slightly improve resilience to future corpus shifts. Not blocking.

Merge Disposition

Merge report at `/home/user/music-gen-instance/fork-675abd086911/clone-1/merge_report.md` (worker-reported; sandbox scope limitation prevents direct verify). Eight shadow-ledger rows queued for `concat_clone_ledgers`; c32-v2 substantive-M-covered convention applies. Root conductor: merge clone-1 shadow ledger; six named events (rubric-frozen $\rightarrow$ verdict-recorded) + two housekeeping (`_archive/cycle-37-scratch-clone-1`, `_infra/adopt-cycle37-tests-clone-1`) all fall under the c32-v2 pattern. The c36 handoff `_manager/ear-sb3-statistic-degeneracy-on-singleton-artists-clone-0` should transition to superseded (by F1_ADOPTED) or validated at merge time.

Cycle-38 Handoff (Pre-Registered Downstream Contract, Rubric §10)

Primary: M-EAR-1/real-label-training-v1 now unblocked in its analytical dimension.

Any M-EAR-1/real-label-training-v1 leak test on the 43-song singleton-artist rated corpus MUST:

1. Call F1 as the statistic (replacing c6's `S = max(S_model, S_resid)` line in `scripts/ear/leak_test.py`; anchor is READ-ONLY under c37's scope, so c38 owns the actual edit).
2. Inspect the singleton-degeneracy invariant (F1 saturates at 2/3 on singleton corpora by construction).
3. Return `SB3_UNRESOLVED_SINGLETON_CORPUS` rather than a numerical detection percentile on the singleton-artist rated corpus.

Corpus scale is the leading candidate variable (per c36 clone-0 close): resolving `SB3_UNRESOLVED_SINGLETON_CORPUS` requires within-artist corpus expansion, not chassis redesign (locked out per c22/c23/c25 anti-patterns).

Fallback statistic backup: F3 ($\Delta = 0.020$ in aggregate) remains available if future corpus shifts change the tiebreak.

Guard-refinement candidate (non-blocking infra): the c33 harness-clone-namespace-guard's endswith check should match `-clone-<digit>+/[^/]+` (i.e. any tail past a clone-suffixed parent) as also “already namespaced”, to avoid the cosmetic double-suffix. Worker archived workaround at `tools/stale/_fix_shadow_clone_ids.py`; a proper fix is lint-level.

Sibling handoffs (from cycle-36 fork closure): - M-GEN-1/palette-driven-batch-v4 (deeper sfizz perturbation, opcode-file rewrite per rule) — c37 branch-B concern, not this branch. - palette-v3 VST3 activation (Dexed-only strict-SMALL tolerance-gate primary; Surge XT bisection deferred).

Cumulative Progress

M-EAR-1 arc analytical branch closed:

Cycle	Milestone	Verdict
c22	synthetic-label-audit	Path A insufficient
c23	head-regularization-audit	anti-pattern locked
c25	feature-representation-audit	anti-pattern locked
c26	_manager/M-EAR-1-path-B-commit	committed; three SBs frozen
c31	armed-harness-fixture-reinforcement	FIXTURE_READY
c36	M-EAR-1/real-label-training-v0	EAR_v0_INSUFFICIENT (first real-label fire)
c37 (this)	_manager/ear-sb3-statistic-degeneracy-fallback-statistic	F1_ADOPTED (SB3 statistic-degeneracy blocker discharged)

Pattern durability: six consecutive cycles of rubric-first pre-registration discipline (c26-c37). Every cycle since c26 has committed a verdict rubric before analysis, with rubric SHA embedded verbatim in verdict JSON and a git-mtime-order test asserting it. Zero after-the-fact rubric edits. Recommend codification into plan-of-record standing practice.

Ledger namespace-guard evolution: c32 fanout-namespace-convention → c33 harness-clone-namespace-guard (writer enforcement) → c36 v2 (extends to substantive M-* families) → **c37 clone-1 surfaces the double-suffix-on-already-namespaced-parents subtle interaction**. Each cycle exposes a tighter case; the convention hardens. No cross-clone id collisions observed since v2 landed.

c29 state-machine lemma respected: peer sub-milestone; ledger topology stays a DAG.

Mechanism scoreboard unchanged: M1/M2/M3/M4 collision-modeling arc remains closed as PARTIAL_BP_UNRESOLVED_SHAPE (c30). No new mechanism candidates opened this cycle. 5 anti-patterns locked; c31 STILL_GAP + c35 A + c8 octave + c11 CLAP/VGGish surface documented.

Rated audio egress: still 403 at *.googlevideo.com per c34 baseline. Analytical + fixture-based work continues unblocked; M-EAR-1 real-label posture is armed-and-fired-with-honest-negative-finding, now with statistical fallback in place for the SB3-degeneracy edge case.

[END OUTPUT]